

✓ PRACTICAL 7 — Regression Analysis

TeachingRatings Dataset

Dataset: Teachers / TeachingRatings dataset

Course: Statistical Foundations of Data Sciences

1. Regression with T-test — Does gender affect teaching evaluation rates?
2. Regression with ANOVA — Does beauty score for instructors differ by age?
3. Correlation using regression — Is teaching evaluation score correlated with beauty score?

✓ 1. Import Libraries

```
import pandas as pd
import numpy as np
import statsmodels.api as sm
import statsmodels.formula.api as smf
from statsmodels.stats.anova import anova_lm
import matplotlib.pyplot as plt
import seaborn as sns

print("Libraries imported successfully.")
```

Libraries imported successfully.

✓ 2. Load the Dataset

This uses the same Google Drive path used in the previous practicals. If your CSV is stored somewhere else, change `file_path`.

```
file_path = '/content/TeachingRatings.csv'

df = pd.read_csv(file_path)

print("Dataset loaded successfully.")
print("Shape:", df.shape)
df.head()
```

Dataset loaded successfully.
Shape: (463, 13)

	rownames	minority	age	gender	credits	beauty	eval	division	native	tenure	students	allst
0	1	yes	36	female	more	0.289916	4.3	upper	yes	yes	24	
1	2	no	59	male	more	-0.737732	4.5	upper	yes	yes	17	
2	3	no	51	male	more	-0.571984	3.7	upper	yes	yes	55	
3	4	no	40	female	more	-0.677963	4.3	upper	yes	yes	40	
4	5	no	31	female	more	1.509794	4.4	upper	yes	yes	42	

Next steps: [Generate code with df](#) [New interactive sheet](#)

```
print("Columns:")
print(df.columns.tolist())

print("\nDataset information:")
df.info()
```

```
Columns:
['rownames', 'minority', 'age', 'gender', 'credits', 'beauty', 'eval', 'division', 'native', 'tenure']

Dataset information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 463 entries, 0 to 462
Data columns (total 13 columns):
#   Column      Non-Null Count  Dtype
---  ---
0   rownames    463 non-null    int64
1   minority    463 non-null    object
2   age         463 non-null    int64
3   gender      463 non-null    object
4   credits     463 non-null    object
5   beauty      463 non-null    float64
6   eval        463 non-null    float64
7   division    463 non-null    object
8   native      463 non-null    object
9   tenure      463 non-null    object
10  students    463 non-null    int64
11  allstudents 463 non-null    int64
12  prof        463 non-null    int64
dtypes: float64(2), int64(5), object(6)
memory usage: 47.2+ KB
```

✓ Q1. Regression with T-Test

Question

Using the teachers rating data set, does gender affect teaching evaluation rates?

We use OLS regression with a binary `female` indicator:

- `eval` = dependent variable
- `female` = independent variable
- female = 1, male = 0

```
# Create a binary gender variable
df['female'] = (df['gender'].str.lower() == 'female').astype(int)

# OLS regression
model_q1 = smf.ols('eval ~ female', data=df).fit()

print(model_q1.summary())
```

Q1 Interpretation

The professor's expected output is approximately:

- Female coefficient = **-0.1680**
- t-statistic = **-3.250**
- p-value = **0.001**
- R-squared = **0.022**

Since $p < 0.05$, reject H_0 .

Conclusion: There is a statistically significant relationship between gender and teaching evaluation rates. In this model, the estimated evaluation score for female instructors is approximately 0.168 points lower than for the male reference group.

✓ Q2. Regression with ANOVA

Question

Using the teachers' rating data set, does beauty score for instructors differ by age?

Age is grouped into three categories:

1. 40 years and younger: $\text{age} \leq 40$
2. Between 40 and 57 years: $40 < \text{age} < 57$
3. 57 years and older: $\text{age} \geq 57$

```
# Create the age groups
df['age_group'] = pd.cut(
    df['age'],
    bins=[-np.inf, 40, 57, np.inf],
    labels=[
        '40 years and younger',
        'between 40 and 57 years',
        '57 years and older'
    ]
)

print(df['age_group'].value_counts().sort_index())
```

```
age_group
40 years and younger    113
between 40 and 57 years  256
57 years and older      94
Name: count, dtype: int64
```

```
# Regression model for ANOVA
model_q2 = smf.ols('beauty ~ C(age_group)', data=df).fit()

# ANOVA table
anova_table = anova_lm(model_q2, typ=2)

# Rename the model row to match the professor's requested output
anova_table.index = [
    'age_group' if str(x) == 'C(age_group)' else x
    for x in anova_table.index
]

print(anova_table)
```

	sum_sq	df	F	PR(>F)
age_group	18.964684	2.0	16.252422	1.513160e-07
Residual	268.383213	460.0	NaN	NaN

Q2 Expected Result

The professor's PDF shows the main ANOVA row as:

	df	sum_sq	mean_sq	F	PR(>F)
age_group	2.0	20.422744	10.211372	17.597559	4.322549e-08

Since $p < 0.05$, reject H_0 .

Conclusion: There is a statistically significant difference in beauty scores among the different age groups.

Q3. Correlation Using Regression

Question

Using the teachers' rating dataset, is teaching evaluation score correlated with beauty score?

We use simple linear regression:

- `eval` = dependent variable
- `beauty` = independent variable

```
# OLS regression
model_q3 = smf.ols('eval ~ beauty', data=df).fit()

print(model_q3.summary())
```

```

=====
                        OLS Regression Results
=====
Dep. Variable:          eval    R-squared:                0.036
Model:                  OLS    Adj. R-squared:           0.034
Method:                 Least Squares    F-statistic:         17.08
Date:                  Sun, 27 Sep 2026    Prob (F-statistic):    4.25e-05
Time:                  06:46:57    Log-Likelihood:       -375.32
No. Observations:      463    AIC:                  754.6
Df Residuals:          461    BIC:                  762.9
Df Model:              1
Covariance Type:       nonrobust
=====
                        coef    std err          t      P>|t|      [0.025    0.975]
-----
Intercept              3.9983      0.025    157.727      0.000        3.948        4.048
beauty                 0.1330      0.032      4.133      0.000        0.070        0.196
=====
Omnibus:                 15.399    Durbin-Watson:           1.410
Prob(Omnibus):            0.000    Jarque-Bera (JB):        16.405
Skew:                    -0.453    Prob(JB):                0.000274
Kurtosis:                 2.831    Cond. No.                 1.27
=====

```

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Q3 Interpretation

The professor's expected output is approximately:

- Beauty coefficient = **0.1330**
- t-statistic = **4.135**
- p-value = **4.25×10^{-5}**
- R-squared = **0.036**

Since $p < 0.05$, reject H_0 .

Conclusion: There is a statistically significant positive relationship between beauty score and teaching evaluation score. The R^2 of about 0.036 means beauty score explains about 3.6% of the variation in teaching evaluation scores in this simple regression.

Double-click (or enter) to edit

```
plt.figure(figsize=(8, 5))
sns.regplot(data=df, x='beauty', y='eval')
plt.title('Teaching Evaluation vs Beauty Score')
plt.xlabel('Beauty Score')
plt.ylabel('Teaching Evaluation Score')
plt.show()
```



